

GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: analytic singular structure plus neural smooth correction

$$G_{\theta}(t, \eta) = \underbrace{E(t, \eta)M(t)R_{\theta}(t, \eta)}_{\text{learned smooth correction}} + \underbrace{B(t) \left(J_0(t, \eta) - \frac{1}{2}E(t, \eta) \right)}_{\text{Heaviside cancellation}} + \underbrace{A(t)G_0(t, \eta)}_{\text{Dirac jump}}.$$

LEARNED SMOOTH CORRECTION

$$E(t, \eta)M(t)R_{\theta}(t, \eta), \quad E(t, \eta) = t\eta(1 - \eta), \quad M(t) = 1 - t.$$

GreenNet learns the smooth correction R_{θ} ; the singular and cancellation structure is built into the kernel form.